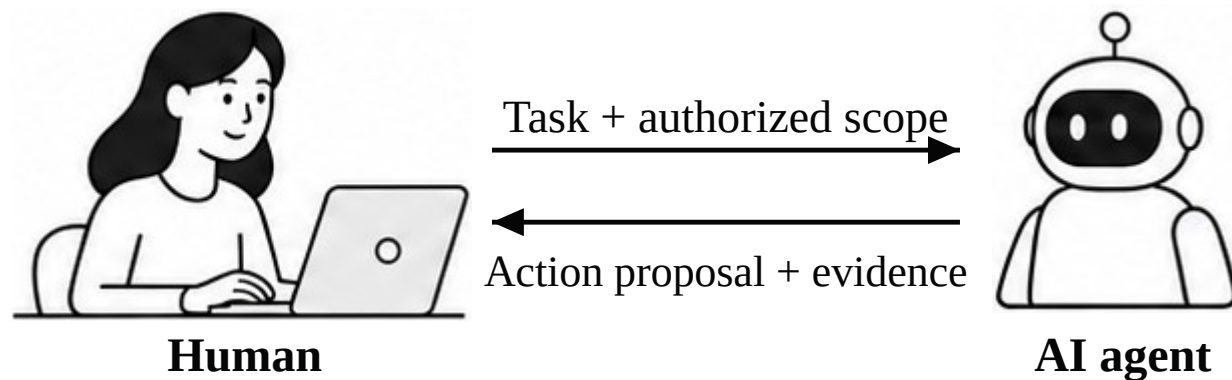


# Reliable AI Agents Under Human Control

A proposed research agenda for actions, system updates, and delegation

## Human–Agent Interaction

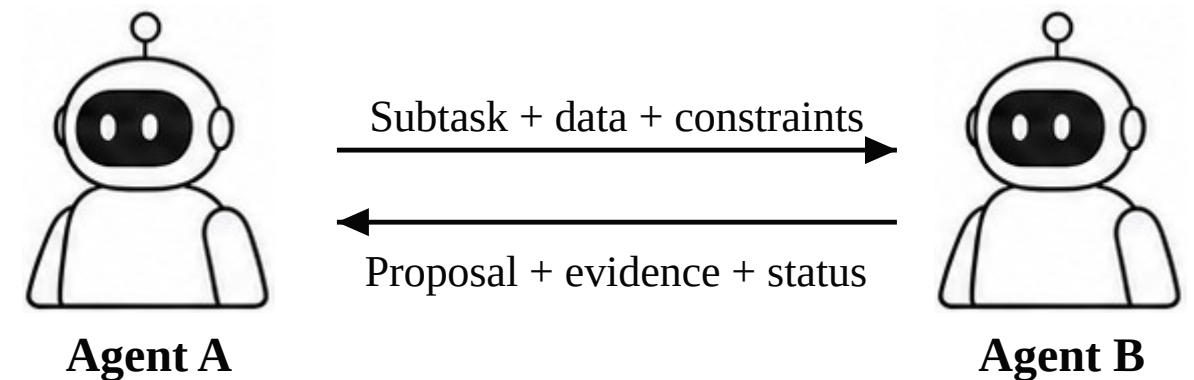


**I** Check proposed actions against approved requirements

**II** Evaluate security and reliability after system updates

**III** Enforce current scope and permissions across task steps

## Agent–Agent Collaboration



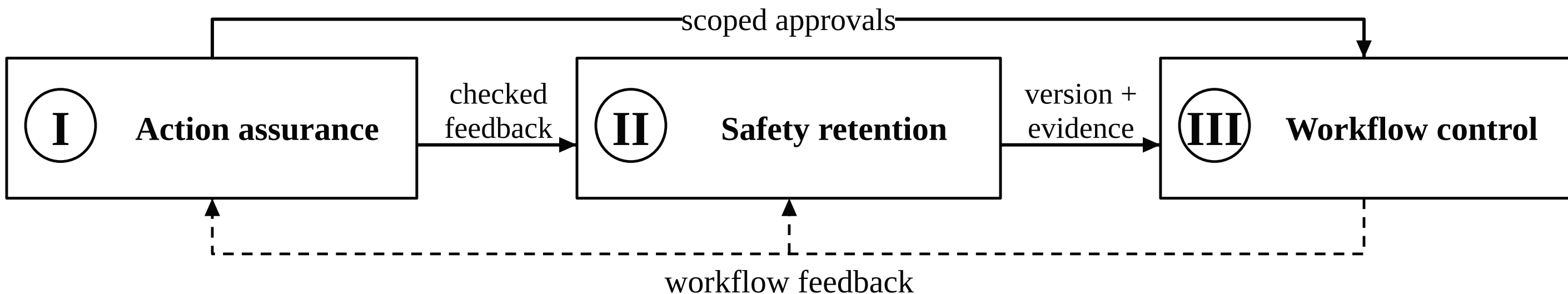
**I** Check proposed actions using independent evidence

**II** Assess retention after either agent or harness updates

**III** Enforce inherited data limits and valid delegation scope

## Shared human-approved requirements $\Phi$

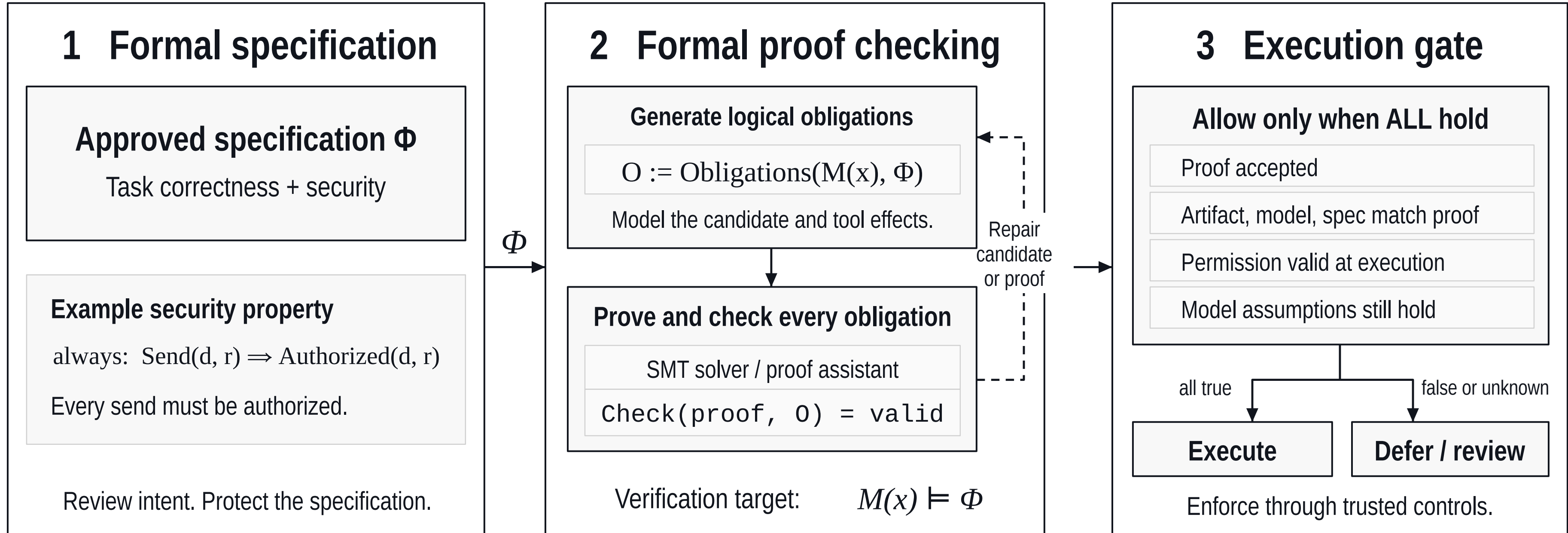
Task criteria · security properties · authorization scope



Formal claims require faithful models, sound checking, and trusted enforcement;  
tests provide evidence for evaluated conditions.

# Formal Oversight Before Action

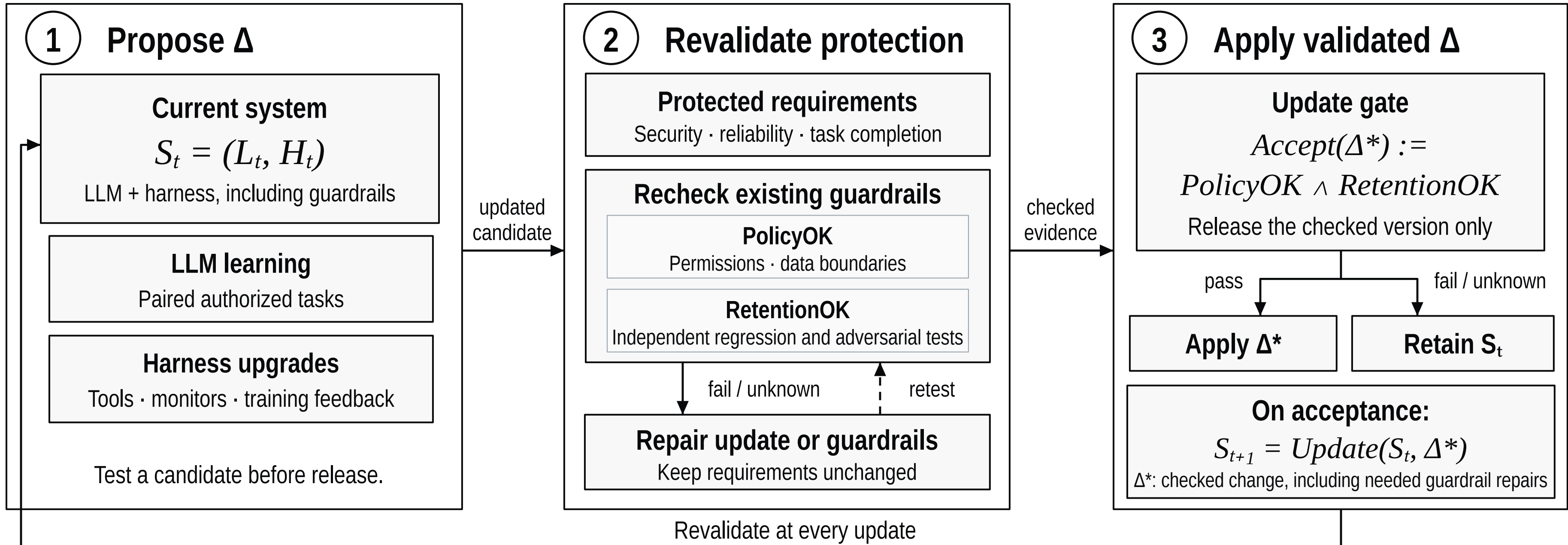
Proposed framework: specify properties, prove obligations, enforce execution



Assurance requires faithful models, sound proof checking, and trusted enforcement.

# Preserving Safety Through Continual Updates

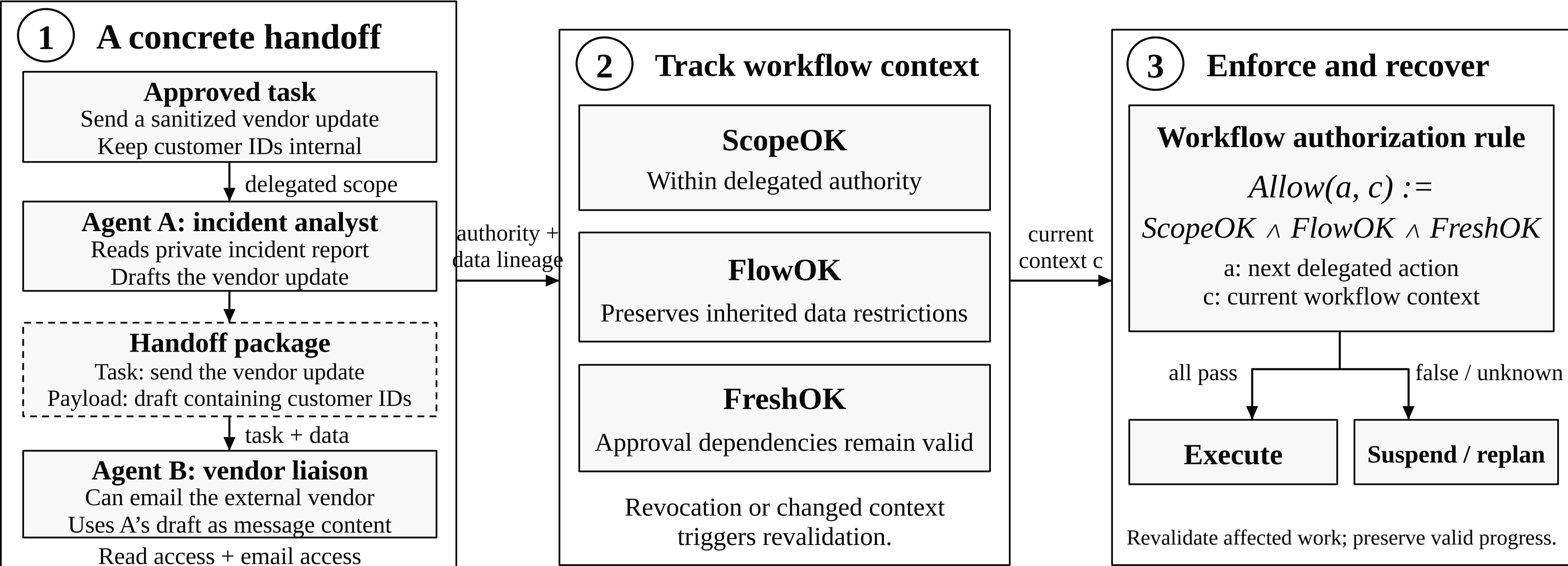
LLM or harness changes can invalidate existing guardrails



Protection requirements stay fixed; guardrail implementations may evolve.

# III. Authorization Across Delegated Workflows

A private report can leak through a delegated email draft.



Guarantees depend on captured effects and trusted enforcement.